Towards an Active In-Vehicle System for Mitigating Vehicle Ramming Attacks

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Abstract. Since the first recorded incident in 1949, vehicle ramming attacks (VRAs) have resulted in over 400 fatalities and more than 2,100 injuries worldwide. A common preventive measure involves the physical separation of roadways and pedestrian areas using Vehicle Security Barriers (VSBs), such as concrete blocks or retractable bollards. While effective, these passive, location-specific solutions cannot fully prevent VRAs, as attacks can occur anywhere. A more comprehensive approach requires an in-vehicle system that travels with the vehicle, ensuring protection regardless of location. In this study, we propose and evaluate an active in-vehicle system designed to mitigate VRAs by restricting the buildup of excessive kinetic energy when malicious intent is suspected. This novel approach offers a dynamic, location-independent solution to counteract VRAs at their onset.

Keywords: vehicle ramming attacks · driver intent detection · vehicle acceleration control · pedestrian safety · threat analysis · system integrity

1 Introduction

Since the first recorded incident in 1949, Vehicle Ramming Attacks (VRAs) have resulted in over 400 fatalities and more than 2,100 injuries worldwide [4]. A standardized [1] preventive measure is physically separating pedestrian areas with Vehicle Security Barriers (VSBs). VSBs cannot effectively prevent VRAs, as attacks can occur anywhere. A location-independent solution can only be designed as an integral part of the vehicle system, so that the vehicle carries the solution together with itself to all places.

There is not much information-technology oriented work to avoid VRAs. For example in reports on swarm intelligence that mention VRAs in their introductory sections, it is not elaborated on how a VRA mitigation specific solution will be achieved by the provided method [2, 3]. These reports might have been written based on a presumption like in a remote shutdown application [4], that

[4] These numbers are collected by combining event information from Wikipedia pages that list VRAs in languages English, German, French

a VRA suspect is known prior to the event and can be tracked. We think a solution, operating without a presumption of a known attacker is the right direction. Not depending on apriori knowledge increases the outreach of the method.

Another report mentions VRA detection among possible applications of the map-matching method [6]. We beleive map matching is a useful method to detect irregular driving paths, however it has the disadvantage, to generate false positive alarms for the VRA detection if the map is outdated.

In this work, we propose an early-stage function concept of an in-vehicle function to avoid VRAs. The function Amok Drive Inhibition (ADI) defines a proactive measure against VRAs, such that if implemented with performance sufficiency, would avoid VRAs via inhibiting unreasonable accelerations at their onset if malicious intent is suspected. ADI function differs from Pedestrian Collision Avoidance (PCA) functions, in that it proactively rejects an excessive acceleration request before a collision risk is materialized, while PCA functions decelerate the vehicle by applying brakes reactively after a collision risk has already materialized.

The main motivation of the raw architecture proposal is to provide an initial item definition seed for public ISO 26262, ISO 21434, and ISO 21448 [17, 18, 16] analyses, and invite the research community to assess, implement, and test the seed idea, to arrive at a mature and socially acceptable concept in an open setting.

In our proposed architecture the focus is put on providing a minimalistic hardware configuration with hardware that comes as a standard feature because of regulations. This abstract function architecture does not define which computing hardware partitioning of the function would be proper. The focus is kept on the logical flow of information in a sense-decide-act pattern. Achieving a performant system under this constraint enables the largest possible adoption of the idea, which in turn maximizes the implied public safety.

2 Main Function - Amok Drive Inhibition

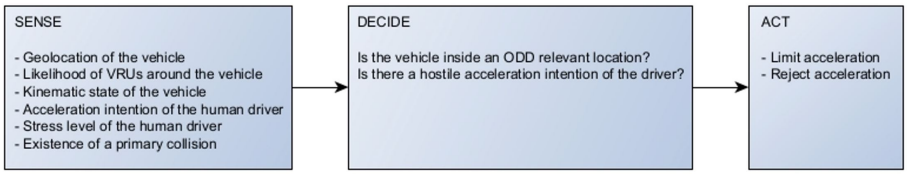

Fig. 1. Sense-Decide-Act view of ADI.

ADI function follows the SENSE-DECIDE-ACT functional chain (figure 1) which is typical of driving functions. The functional chain starts with the collection and preprocessing of data for use in the decide part: 1) The kinematic

state of the vehicle to validate the GNSS coordinates, 2) Road section characteristics like typical speed 3) Presence of Vulnerable Road Users (VRUs) in the vicinity of the vehicle, 4) The acceleration pedal behavior of the human driver, 5) The health and emotional state of the driver, and 6) Collision sensor input to mitigate secondary collisions as a fallback mechanism.

The decide part will assess whether the vehicle is within the Operational Design Domain (ODD) of ADI function, and if within ODD, assess whether the driver has a hostile intent. If a hostile intent is concluded, the decide part will send an acceleration inhibit request to the act part.

ADI propagates the excess acceleration request generation in two paths. In the main path, the subcomponent Ramming Intention Assessment (RIA) generates an estimation of the malicious intent which leads to a request to inhibit excess acceleration. For the case where the RIA function underperforms and fails to detect a VRA intent, a fallback path is proposed. The second path serves as a fallback level to avoid secondary collisions whereas the first path shows performance insufficiencies to detect a VRA. The second path reads the presence of a Primary Collision Detected (PCD) signal to override the RIA and make a direct permanent acceleration inhibition request to avoid secondary collisions.

The act part of the functional chain implements the inhibition action within existing physical systems that transmit torque to the wheels (2.4).

A simplified overview of the function architecture can be seen in the figure 2.

2.1 ODD Check

ADI has a 2 dimensional ODD. The resulting alertness level can be set as a multiplication of the enumerated criticalities of the 2 ODD dimensions.

The first ODD dimension of ADI is the class of the current road section on which the vehicle is driving. Road sections with confrontation surfaces with pedestrians are the region of interest for the ADI function and the ODD is constrained accordingly. The reliability of GNSS information is a fundamental requirement for the ADI function. Preprocessing function Real Time Kinematics (RTK) will validate the raw GNSS information before it is fed into an HD Map and detect if a GNSS Spoofing cyberattack is happening. Also, radio frequency level detection methods can be used, but we assume that these are not ADI specific requirements.

Recovery from GNSS spoofing For ADI, GNSS spoofing detection is necessary but not sufficient for a healthy operation. Data engineering methods can be used to not only detect but also to recover from a GNSS spoofing attack [20]. If a GNSS spoofing is detected but cannot be mitigated, the ODD Check must report 'ouf of ODD' and render the ADI dysfunctional. Another possible solution to recover from a detected but not mitigated GNSS spoofing attack could be to drive a location hint based on the access points, and mobile data for location analysis. [7] shows that knowing a small number of trusted wifi access points with

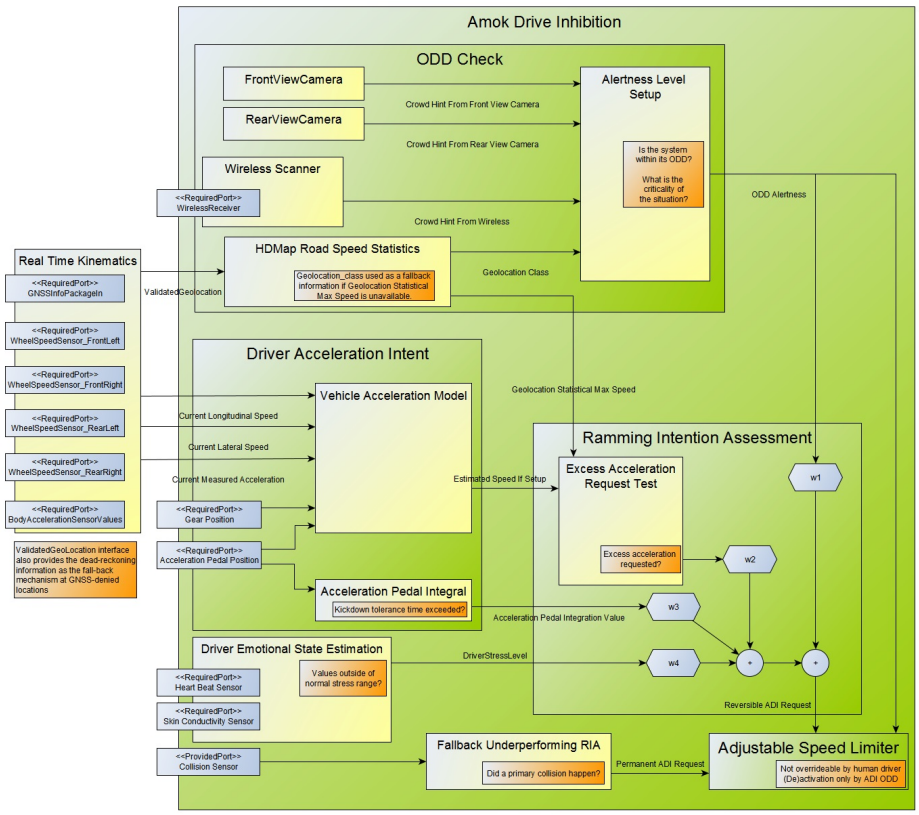

Fig. 2. Overview of ADI function architecture.

known geolocations installed in a city, it is possible to derive the geolocation of a moving receiver precisely.

The second dimension of ADI ODD is the presence of crowds in the vicinity of the ego vehicle. A presence of crowds around the vehicle can be assessed by cameras and wireless device scanning methods. The advantages and disadvantages of both methods will be discussed in the section 2.1. We suggest a combined use of both methods in a complementary way.

Speed Category Check If the geolocation is validated to be spoofing free, it is fed into an HD Map with road section speed category information. HD map delivers the typical speed range category of the road section.

Table 1. HD Map Road Classes and example alertness levels

Road Class	Description
0	Motorways or freeways with controlled access
1	Major roads, typically primary highways
2	Secondary roads, often significant regional connectors
3	Local connecting roads
4	Local roads
5	Ramps or special segments connecting different classes
6	Unpaved roads
7	Parking lot roads
8	Private roads
9	Walkways or pedestrian paths
10	Alleyways
11	Driveways
12	Emergency vehicle access roads

If the speed category is unknown for a road section, road section type (see table 1) can be used as a fall-back solution. In the fallback case, a mapping from road section types onto speed categories must be defined. An example mapping for the fallback solution can be seen in the table 2. Speed categories 4 and above can be taken as a seed ODD constraint to explore the function field in experiments.

Crowd Detection As most VRA attacks happen as frontal hits, on the surface it looks like relying only on the front camera system would be sufficient to

Table 2. Road Class - Speed Category Mapping

Speed Category	Road Class	ADI Alertness level	Speed Range in km/h
1	0	0	> 130 km/h
2	1	0	101-130 km/h
3	2	0	91-100 km/h
4	3, 4	1	71-90 km/h
5	5	2	51-70 km/h
6	6	3	31-50 km/h
7	7, 10	4	11-30 km/h
8	8, 9, 11, 12	4	< 11 km/h

perform a crowd check. Unlike in an automated or peaceful manual driving scenario, in an amok drive, the trajectory can abruptly change and does not follow conventional driving behavior patterns. Therefore a check for the existence of a crowd in the vicinity of the vehicle can happen omnidirectional.

The exact location of the individuals and trajectory analysis will not be used, because the ADI function will not monitor a collision risk, but a VRA intention of the human driver. The information on the existence of crowds nearby will take part in setting the ODD-in-scope state for the VRA intention assessment.

VRA scenarios resemble a negative version of the moral dilemma problem stated in [11], where the amok driver aims to choose a scenario that causes the most harm. The utilitarian solutions of the moral dilemma scenarios face criticism that the systems should not be designed to prioritize characteristics of the potential victims [10]. In [12] Awad et.al. suggest that not prioritizing the options leaves the problem open, and a resolution based on survey data gives a practical answer to the questions of social acceptability of harm. They also defend the opinion that such pragmatic priority resolution makes the system behavior explainable and defendable after a harmful event.

The question of which sizes of a crowd must contribute a higher level of system alertness poses an ethical question. This ethical question is by definition different than the A or B questions defined in [11]. We believe omnidirectional crowd detection methods make the question of how to prioritize VRU characteristics irrelevant. Camera based crowd detection output must be only a number so that the ethical question also drops there.

Camera based Crowd Check In Europe, cameras belong to standard vehicle configuration as a result of the Regulation (EU) 2019/2144 on road vehicle type approval, of which the first stage entered into force in July 2022 [13]. Within EU Regulation 2019/2144, especially Emergency Lane Keeping System ELKS and Reversing Detection System RDS make it necessary that front and rear cameras are installed in all road vehicles.

A weak point of the camera based crowd check is the fixed field of view of a camera. In a crowded lowest speed pedestrian area, a driver with a VRA intent can turn his head around to look for a bigger crowd to cause bigger harm. During the driver's maneuver towards a bigger crowd, the camera can temporarily point towards areas without people. This problem is likely to happen in the lowest speed road sections of categories 7 and 8 (see table **??**). For Crowd Check at the lowest speed areas a different modality is needed to augment the Camera based Crowd Check.

Wireless Device Scan based Crowd Check In the lowest speed pedestrian areas, the hostile driver can switch gears back and forth to make significant changes to the direction, in his search for the bigger crowd. For such scenarios, an omnidirectional crowd detection method is necessary. Even though surround view multi camera systems can provide the necessary omnidirectionality, these systems are not a standard configuration enforced by a regulation at the time of this writing. Therefore, a large scale adoption of surround view multicamera systems cannot be assumed.

Under the assumption that the number of mobile phones around can be a good approximation to the number of people, wireless device scans can be a possible augmenting method for Crowd check. Omnidirectionality in wireless device scanning addresses the uncertain nature of the next action direction of a hostile driver.

To be useful for crowd detection, a wireless technology must have become widely used in smartphones. This makes Bluetooth Low Energy (BLE) and Wi-Fi technologies the only possibilities. Ultra Wide Band (UWB) technology is emerging in a few smartphone models but is currently far from being in wide use. Experiment results in [5] suggest that BLE is not a suitable technology for crowd detection even in a stationary setting. Their data also shows that WiFi scanning can be useful in crowd detection in a stationary setting. A simplistic check with a mobile phone WLAN scanning suggests that access points outside a moving car can be detected, even at high speeds. In this simple check, we could see that even at 130km/h ego vehicle speed access points of the cars from the other motorway direction can be detected[5]. This suggests that the low refresh rate of the scan is due to the filtering of the strongest signals, not the speed of travel. The feasibility of WiFi scanning for ADI must be tested with a proper experiment setup that can handle lower level WiFi signals without filtering.

In the VRA in December 2024 in Magdeburg [19], the attacker approached the Christmas Market that was protected by VSBs through the emergency vehicle corridor at a slow speed (VSBs were ineffective). At such slow speed approaching drives, wireless device scans can provide a crowd approximation with its refresh rate that can be useful for the ADI use case.

[5] Brand names in the default access point names were compared to the vehicles on the same side of the motorway.

2.2 Driver Acceleration Intent

Driver Acceleration Intent component provides an estimate of the future speed of the vehicle if the driver request is realized and the integration value of short term acceleration pedal values.

Acceleration Pedal Integral receives and accumulates the driver's inputs from the acceleration pedal position sensor. The accumulation of the acceleration pedal inputs over time is a central source of information in the detection of the VRA intent of the driver. If the pedal kickdown level and its duration exceed normal operational ranges, a VRA intent is suspected. This base information will be supported and validated by additional information in the next stations of data processing.

Vehicle Acceleration Model (VAM) estimates the future speed and acceleration of the vehicle using a kinetic model of the vehicle. VAM estimates the calculated future speed of the vehicle if the acceleration pedal input is processed without any acceleration inhibition. If the future speed estimation exceeds the road section speed levels significantly, the malicious intent of the driver is suspected. This information will support malicious intent detection.

A time distance parameter for the time horizon of future speeds must be defined to test the effect of different parameter values on the reliability of intent assessment in field tests and simulations. As it is impractical to collect experimental data on the behavior of a hostile driver, that behavior must be made simulation ready by definition of reinforcement learning agents, with their reward function is a maximization of the harm caused.

2.3 Ramming Intention Assessment (RIA)

Ramming Intention Assessment (RIA) is the central ADI component that reads all the prepared information for an assessment of a VRA intent of the human driver. RIA mainly assesses the acceleration pedal input of the driver against tolerable levels of kickdown accumulation (percentage x seconds) depending on the vehicle type. The assessment is supported with additional hints about the relevance of malicious intent: signs of extreme stress at the driver's hands, alertness level of the geolocation, and a crowd hint.

Time aspect of input cues Onset times of different decision-making input cues are not time correlated and their refresh rates differ. For this reason, high alertness levels of the input cues must be held concerning the refresh rate of the slowest element.

Based on an additive score for VRA intent added from the input cues, the acceleration inhibition requests can be modeled to be a graceful, progressive inhibition, such that any false positives do not immediately lead to a complete traction loss.

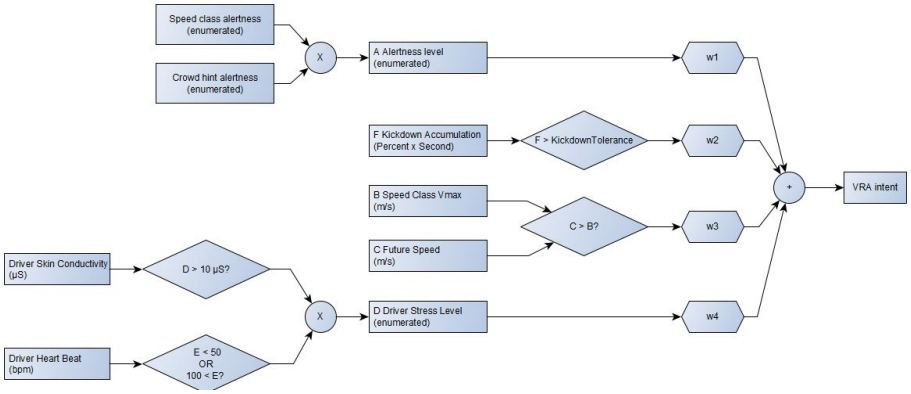

Fig. 3. Controllability question in a VRA situation.

Excess Acceleration Request Test Accumulation of the strength and duration of the acceleration pedal position gives a measure of the driver's intention. If the pedal is kicked down longer than normal operational conditions There exist legitimate operational scenarios for an acceleration pedal kickdown like forcing a gear switch in an automatic gear ICEV. There may also be legitimate scenarios to drive around a car stopped in the lane while a counter traffic is approaching at a distance. However, after all these scenarios played out, there will be an upper limit of legitimate use of an acceleration pedal kickdown.

Depending on the vehicle's acceleration behavior and vehicle technology, kickdown tolerance time must be adjusted. ADI has three variants based on vehicle drive train technology:

- VM1.M - Manual Transmission Internal Combustion Engine Vehicles (MT-ICEV): Kickdown tolerance time should be constrained by the vehicle acceleration behavior.
- VM1.A - Automatic Transmission Internal Combustion Engine Vehicles (AT-ICEV): Kickdown tolerance time shall be extended by the legitimate kickdown duration required to enforce a gear transition of the transmission system.
- VM1.E - Battery Electric Vehicles (BEV): Kickdown tolerance time shall be much stricter compared to the Manual Transmission ICEVs, because of the higher acceleration capabilities of BEVs.

An example setting of tolerance time can make use of a look-up-table (LUT) like in figure 4.

Driver Emotional State Estimation Driver emotional state can be estimated by a camera based driver monitoring system. However, a hostile driver can easily disable a camera based driver monitoring system using a sticker. With this disadvantage of camera based driver monitoring, we propose the use of simpler

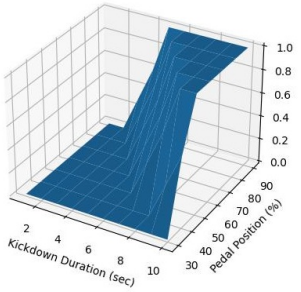

Fig. 4. Example acceleration pedal tolerance LUT for experiments.

sensors on the steering wheel. Steering wheel touch sensors like heartbeat sensors and Galvanic Skin Response (GSR) sensors can be used to monitor the changes to the driver's emotional state. Raw sensor levels associated to panic and fight-or-flight levels of high stress situations are used as a supportive cue in RIA. The question can arise if the steering wheel hands–on sensors manipulation by water bottles, and oranges stretched between spikes of the steering wheel is not a disadvantage. In an amok drive the person sitting at the driver's seat is not interested in keeping hands off the steering wheel on the contrary, attaches tightly to the steering wheel to achieve the malicious intent.

Fallback Path for Underperforming RIA Systems like Secondary Collision Mitigation Braking (SCMB), and Airbag Control Unit (ACU) rely on input from crash sensors to engage. These systems are limited to high speed crashes as they trigger airbag engagement. A trivial evidence for this limitation is the fact that hostile drivers can to continue crashing into more people after their first hit without having the airbags opened [19]. Also, SCMB does not get engaged, or is ineffective, as the vehicle does not come to a stop after the first collision. This limitation is by design to avoid crash sensor false positives that would cause hazardous airbag engagements at high speeds. To avoid false positives, the sensitivity of the crash sensors used for SCMB and ACU is optimized for high speed crashes.

To keep the sensitivity of the airbag crash sensor untouched while providing useful crash detection input for ADI, a separate crash sensor path must be used with optimized sensitivity for speed ranges in ADI ODD. If experiments show that the state of the art acceleration sensors have insufficient sensitivity, multimodal sensing can be used with pressure sensors under the VRU head impact zone.

If torque inhibition is requested by the fallback path, Excess Torque Inhibition must be implemented as a permanent, irreversible mode switch. Permanent ETI mode must not be reset with a Motor Start/Stop Cycle (MSSC), as in that case, the hostile driver could continue ramming simply by going through

an MSSC. Instead, a mode reset must be made only possible after stopping and the driver mounting off the car. The state transitions can be seen in figure 5.

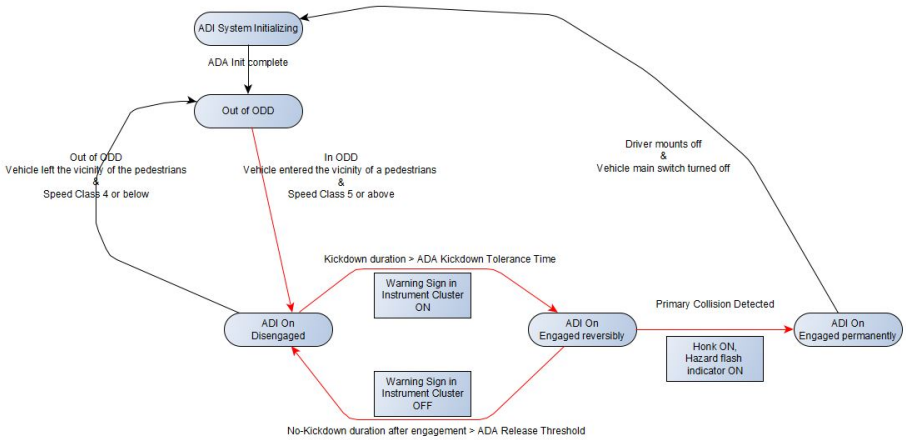

Fig. 5. State Model Overview of ADI.

2.4 Excess Torque Inhibition

Excess Torque Inhibition (ETI) component receives Amok Drive Inhibition Requests and implements the inhibition within the physical system. ETI must manipulate physical torque generating systems which differ by vehicle power train types. The method to inhibit the excess acceleration request of the driver differs between MT-ICEVs, AT-ICEVs, and EVs. The technologies used to address UNECE regulation R89 Adjustable Speed Limitation Device (ASLD) [15] can be used with only change to the requirement 5.2.7.1: "The ASLF must be capable of being activated/de-activated at any time". For ADI to be effective, the speed limiting must not be capable of being deactivated by the driver. Activation/Deactivation of the speed limiter must happen according to the ADI ODD.

As these physical systems do not sit inside the same computing and control units as the host control unit of RID, a secure data communication on the request channel is vital for the integrity of the ADI function. Furthermore, the target ECU for ETI must be mounted on a hard to reach location inside the engine/electric motor/gear box housing to make brute force attacks harder to perform.

3 Discussion

The proposed ADI function poses interesting research questions towards established methods related to road vehicle safety. In safety engineering, the controllability aspect is a fundamental part of the risk assessment of a particular function. The controllability question is implicitly assuming that the human driver is sane and willing to avoid a crash situation. In the ADI use case, however, the human driver is the cause of the hazard and hence the controllability aspect is turned around between the car and the human driver: Does the road vehicle have the controllability of the situation once the malicious intention of the human driver is detected?

With the deliberate intention of the hostile human driver to cause harm, the severity aspect of the hazardous event is at its maximum. The safety analyst may fall into the illusion, that the exposure aspect of the hazardous event being low, looking at the number of VRA events since 1949. This illusion might be a reason why a comparable function like ADI does not exist in production cars. We believe ADI exposure must be treated as the likelihood of harm given an amok driver is driving.

The most comparable function regulated by European regulation [13] to become a standard feature in passenger cars is Intelligent Speed Assist (ISA) [14]. Even though ISA defines a Speed Control Function (SCF), it is overrideable and can be deactivated suggesting a prioritization of comfort over safety. Furthermore, vehicle manufacturers can choose not to implement SCF and keep ISA a warning-only function. This way, ISA fails to provide any mitigating effects against VRAs. However, ADI is compatible with ISA that implements an SCF.

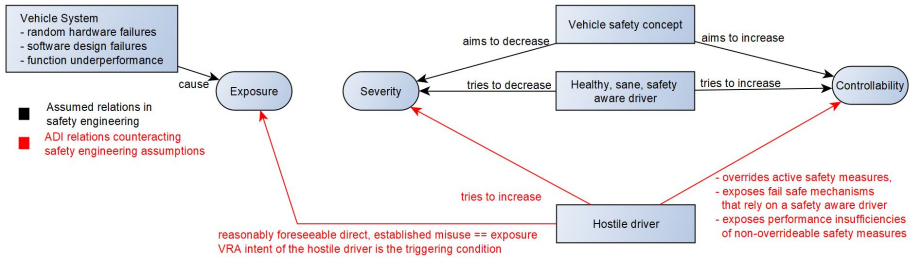

Fig. 6. Controllability question in a VRA situation.

A hostile driver does not only make system controllability assumptions invalid, a hostile driver also deliberately overrides safety mechanisms and exposes performance insufficiencies in them. SOTIF [16] Term 3.17-'misuse' excludes the intention to cause harm and closes the door to the analysis of why secondary collisions are not being avoided in a VRA event and why VRAs are still possible with new cars. We suggest a deliberate intention to cause harm can be characterized as a reasonably foreseeable direct and established misuse.

4 Conclusions

To arrive at a solution suitable for a ubiquitous adoption of ADI, we focused on the definition of a minimalistic, in-vehicle solution for ADI avoiding any hardware that is not available in a car with standard features.

ADI case provides a contradiction to the assumption of a human driver taking good care of critical situations with a safety aware attitude. We believe that among the strong contrasting states of an amok driver and a fully awake, healthy, sane and prompt driver there may be different states of awareness and care. This perspective provides an opportunity to discuss the sufficiency of the ISO 26262 [17] exposure, severity, and controllability matrices that are also used in ISO 21448 [16].

Disclosure of Interests. ADI idea emerged after the frequent VRA attacks in the years 2015–2016 and was iterated on occasionally by the first author. Other authors have contributed some parts of the writing as part of a safety engineering exercise. On our side, there have been and will be no project budgets to further develop the idea with the engineering rigor it requires. With this in mind, this report, in its low maturity of implementation, is now made public as a social responsibility act to enable the safety research community to iterate on and further develop ADI. No patents pending.

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